

COLUMBIA, MO, USAWMO: **724450**Lat: **38.817N**Lon: **92.218W**Elev: **893**StdP: **14.23**Time Zone: **-6.00 (NAC)**Period: **94-19**WBAN: **03945****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.2	8.9	-5.2	4.3	5.1	-0.7	5.5	10.8	26.9	38.8	24.8	36.6	8.5	310	0.519

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	94.7	75.7	91.9	75.9	89.2	75.0	79.5	89.5	78.0	87.9	76.6	86.1	9.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.6	143.2	85.7	75.1	136.1	84.2	73.6	129.4	82.2	43.6	89.6	42.0	88.0	40.5	86.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.0	20.5	18.5	DB	-3.0	99.1	4.7	3.9	-6.4	101.9	-9.1	104.1	-11.8	106.3	-15.2	109.1	(4)
(5)			WB	-3.7	81.4	4.7	1.7	-7.1	82.6	-9.8	83.6	-12.4	84.6	-15.8	85.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.4	30.0	34.8	45.1	55.8	65.0	73.7	77.6	76.5	68.7	57.0	45.0	34.4	(6)	
	DBStd	18.92	12.06	11.91	11.34	9.45	8.00	6.00	5.44	5.52	7.47	9.22	9.99	10.77	(7)	
	HDD50	2093	625	443	231	48	3	0	0	0	1	38	209	495	(8)	
	HDD65	4843	1084	847	622	301	103	9	0	1	46	279	601	950	(9)	
	CDD50	4062	6	16	78	221	466	712	857	822	562	253	59	10	(10)	
	CDD65	1341	0	0	4	24	102	271	392	359	158	30	1	0	(11)	
	CDH74	12504	0	1	39	231	811	2450	3989	3376	1353	243	11	0	(12)	
	CDH80	4964	0	0	2	40	205	920	1776	1496	470	55	0	0	(13)	
	WSAvg	9.2	10.4	10.4	10.9	11.1	9.1	8.2	7.1	7.0	7.6	9.0	10.0	9.9	(14)	
	PrecAvg	39.8	1.8	2.0	3.0	4.4	4.8	4.1	3.9	3.8	3.7	3.3	2.8	2.3	(15)	
(16) Precipitation	PrecMax	62.6	5.9	6.2	10.1	11.7	12.3	10.3	12.1	14.9	12.0	11.0	10.4	7.0	(16)	
	PrecMin	23.7	0.0	0.1	0.7	0.4	1.3	0.4	0.2	0.2	0.3	0.4	0.4	0.2	(17)	
	PrecStd	8.6	1.2	1.3	1.7	2.3	2.3	2.3	2.7	2.9	2.8	2.2	2.1	1.5	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.3	71.0	78.4	84.4	88.5	94.6	100.9	99.6	92.8	85.7	75.4	66.4	(19)	
		MCWB	55.6	57.8	62.4	66.1	72.5	75.1	74.8	75.1	73.9	71.1	62.9	59.3	(20)	
	2%	DB	58.5	64.2	73.5	80.2	85.0	90.7	95.0	94.9	89.0	80.4	69.5	60.4	(21)	
		MCWB	50.9	54.3	59.0	63.3	71.0	75.3	76.9	75.1	73.3	66.0	59.0	54.0	(22)	
	5%	DB	53.1	59.0	68.1	76.1	82.1	88.2	92.0	91.7	85.6	75.9	65.0	55.0	(23)	
		MCWB	46.4	50.7	56.6	61.7	69.6	74.6	77.0	75.3	71.7	64.1	56.5	48.8	(24)	
	10%	DB	47.1	52.9	63.0	72.1	78.9	85.6	89.0	88.2	81.9	71.9	60.5	50.1	(25)	
		MCWB	41.7	46.0	53.8	59.8	67.6	73.5	76.0	74.4	69.6	62.7	53.2	44.6	(26)	
	0.4%	WB	58.5	60.7	65.0	68.9	75.4	79.1	81.5	81.2	77.3	72.7	65.0	60.8	(27)	
		MCDB	64.2	66.6	75.6	79.9	85.3	89.0	90.5	90.9	88.3	82.3	72.4	65.7	(28)	
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	52.7	56.3	61.5	66.1	73.3	77.4	79.9	79.2	75.2	69.6	61.3	55.4	(29)	
		MCDB	56.3	61.9	70.0	76.1	82.5	87.2	89.8	89.1	85.2	76.2	66.6	59.8	(30)	
	5%	WB	46.8	51.3	58.5	63.9	71.1	75.9	78.4	77.4	73.5	66.9	58.1	50.1	(31)	
		MCDB	52.3	58.3	66.2	73.0	79.6	85.2	88.4	87.1	81.8	72.9	63.4	53.8	(32)	
	10%	WB	41.6	45.9	54.6	61.5	69.1	74.6	77.2	75.6	71.7	64.0	54.4	45.2	(33)	
		MCDB	46.8	53.3	61.8	70.1	76.3	83.6	86.7	84.8	79.3	70.8	59.9	49.5	(34)	
	5% DB	MDBR	16.8	17.9	19.8	20.6	19.2	18.7	19.2	20.0	20.8	20.1	18.3	16.0	(35)	
		MCDBR	25.0	26.8	26.1	25.7	21.9	21.1	22.3	24.1	23.0	24.1	24.2	22.9	(36)	
		MCWBR	18.0	18.5	14.9	14.2	11.4	9.5	8.3	8.9	10.0	12.6	16.2	17.0	(37)	
		MCWBR	22.6	24.9	22.6	22.2	20.0	19.3	19.5	20.4	19.7	19.7	21.1	20.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.297	0.312	0.340	0.386	0.429	0.428	0.447	0.452	0.386	0.345	0.304	0.293	(40)	
		taud	2.466	2.418	2.407	2.283	2.225	2.266	2.214	2.214	2.364	2.472	2.529	2.499	(41)	
	Ebn at Noon		280	290	292	282	271	271	264	260	272	274	276	273	(42)	
		Edh at Noon	25	30	34	41	44	43	45	43	35	29	24	23	(43)	
(44) All-Sky Solar Radiation	RadAvg		670	899	1199	1572	1791	2033	2005	1767	1527	1077	740	574	(44)	
	RadStd		77	93	108	103	155	169	130	97	99	118	75	60	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+250	+158
(47) Regional (21 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+148

Nomenclature: See separate page